PROJECT 05

Oil Dripping Tray — 3D Model Redesign

Effective Engineering (Client: Laugfs Lubricant Co.)
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PROJECT SUMMARY

Redesigned the 3D CAD model of an oil drip tray mechanism for a Laugfs lubricant filling machine. The original tray geometry was wrong — oil was dripping on the factory floor instead of being caught and returned to the tank. The updated model fixed tray position, drain slope, and bracket geometry, and new fabrication drawings were issued.

OBJECTIVES

- Fix tray position to within ±5 mm directly under the nozzle drip point.
- Set drain channel slope to at least 3° for gravity oil return.
- Ensure all parts fit within the existing machine envelope.
- Issue revised fabrication drawings for the workshop.

HOW IT WORKS

After each fill cycle, a PLC signal extends a pneumatic cylinder to move the tray under the nozzle. Drip oil falls into the tray, drains via a sloped channel, and returns to the main oil tank. When the next fill starts, the cylinder retracts the tray. The redesign covered the tray shape, drain slope, and the mounting bracket connecting the tray to the cylinder rod.

KEY COMPONENTS

Component	Specification / Function
Oil Drip Tray Pan	Redesigned geometry — correct nozzle coverage
Drain Channel	Minimum 3° slope — gravity oil return
Mounting Bracket	Redesigned to match actual cylinder rod interface
Return Oil Line	Flexible tube back to main oil tank
Pneumatic Cylinder	Existing (unchanged) — moves tray in and out

TECHNICAL SPECIFICATIONS

Parameter	Value	Notes
Tray position target	Within ±5 mm of nozzle drip point	—
Drain slope	≥ 3°	Gravity-fed return
Cylinder stroke	[TO BE FILLED]	Existing actuator, unchanged
CAD tool	SolidWorks	Model redesign + drawings

MY CONTRIBUTION

- Measured installed machine dimensions with caliper to get correct dimensions.
- Identified geometry errors in the existing SolidWorks model.
- Redesigned tray, drain channel, and bracket in three revision rounds.
- Ran virtual fit checks in SolidWorks assembly to confirm no clashes.
- Issued revised fabrication drawing package to the workshop.

CHALLENGES & SOLUTIONS

The tray had to deploy at a slight angle due to space constraints. The drain slope calculation had to be checked relative to gravity in the deployed orientation, not relative to the tray body — handled in SolidWorks measure tool.

RESULTS

Revised model and fabrication drawings accepted by workshop. Design eliminates floor oil spillage and returns drip oil to the tank automatically. Key lesson: always measure the real machine before starting a CAD redesign.

TOOLS & SOFTWARE USED

Tool / Software	Used For
SolidWorks	3D model redesign, fit checks, fabrication drawings
Vernier Caliper	Site measurements of the installed machine

IMAGES & TECHNICAL DRAWINGS

Laugfs oil filling machine — installation at site